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Analog Electronics Exercises

session 2 solutions

Contact: Xuepu Wu, M.Sc.
Xuepu.Wu@etrovub.be

Solution

Exercise 1

a) The g_{ds} of transistors are neglected; thus, output impedance values of the first and second stages are R_1 and R_L, respectively.
$A_{v1} = -g_{m1} * R_1 = -2m * 5k = -10$ and $A_{v2} = -g_{m2} * R_L = -20m * 500 = -10$. So, $A_v = A_{v1} * A_{v2} = 100 = 40dB$

b) As illustrated in Fig. 3, when C_{gd} has a small value, its contribution to the pole frequency becomes negligible compared to the loading capacitor at the drain node. Since, in our case, C_{gd} of the Q_2 transistor has a much smaller value than C_L, we can neglect C_{gd} for calculating the pole at node 3. So, the total capacitance value is approximately equal to C_L at node 3. Because of the Miller effect, we have the capacitance $C_{gd2} * (1 + g_{m2} * R_L)$ at node 2. In addition, we have the capacitor C_{gs2} in the same node. Again, C_{ds1} has a smaller value compared to $C_{gd2} * (1 + g_{m2} * R_L) + C_{gs2}$; hence, we can neglect it at node 2, as well.
Finally, the total capacitance value is $C_{gd1} * (1 + g_{m1} * R_1) + C_{gs1}$ at node 1. Due to Miller effect, C_{gd1} is multiplied by the gain of the first stage.

Poles:
$|p_1| = w_{p1} = |\frac{-1}{R_S * (C_{gs1} + C_{gd1} * (1 + g_{m1} * R_1))}| = \frac{1}{500 * 130f} = 15.4Grad/s$
$|p_2| = w_{p2} = |\frac{-1}{R_1 * (C_{gs2} + C_{gd2} * (1 + g_{m2} * R_L))}| = \frac{1}{5k * 195f} = 1Grad/s$
$|p_3| = w_{p3} = |\frac{-1}{R_L * C_L}| = \frac{1}{500 * 1.1pF} = 1.8Grad/s$

c) Lets assume that w_T is higher than w_{p1} and w_{p2} (since they are close to each other), and lower than w_{p3}.

$40 - 20 * log(\frac{1.8G}{1G}) = 34.9dB$
$34.9 = 40 * log(\frac{w_T}{1.8G}) => w_T = 13.4Grad/s$
Our first assumption is correct.

Total phase shift$= tan^{-1}\frac{13.4G}{1G} + tan^{-1}\frac{13.4G}{1.8G} + tan^{-1}\frac{13.4G}{15.4G} = 209.1^o$

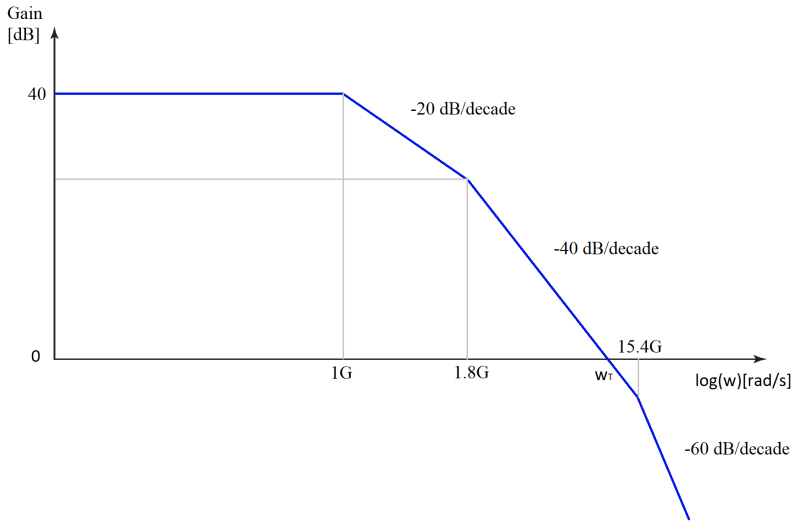


Figure 1: Bode plot

Phase margin $= 180^o - 209.1^o = -29.1^o$

Exercise 2

a) Having a uniform $-20dB/decade$ gain rolloff before w_T means there is only one pole whose value is lower than w_T (by assuming that zeros are far away from w_T).
$40 = 20 * log(\frac{w_T}{w_{p2}}) => w_{p2} = 20Mrad/s$
After adding, w_{p2} will be shifted to lower frequency, and w_{p3} will be shifted to higher frequency. So, w_{p2} will be dominant pole.
$|p_d| = w_{p2} = \frac{1}{R_1*(C_c*g_{m2}*R_L)} => C_c = 1pF$

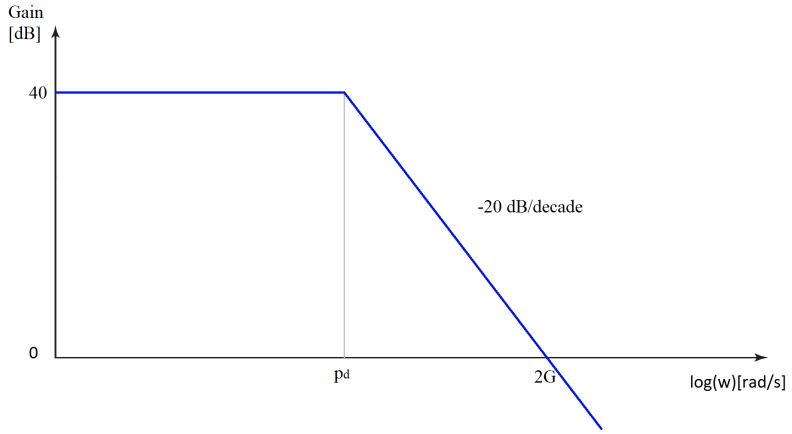


Figure 2: Bode plot

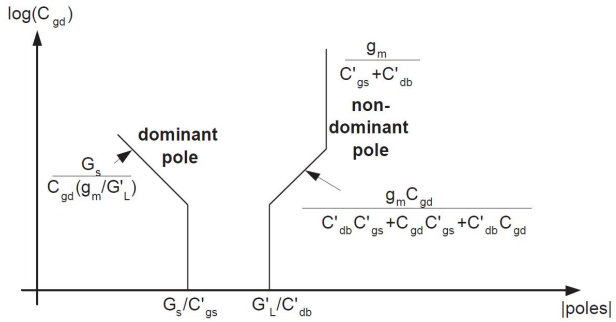


Figure 3: Bode diagram of pole splitting: asymptotes indicate the position of poles as a function of C_{gd} on a log-log scale.

b) After adding such a big capacitor between the gate and drain of the Q_2, the

gate and drain of Q_2 will be shorted at high frequencies. So, the impedance seen from the drain of Q_2 will be equal to $1/g_{m2}$, which is much smaller than R_L. Thus, the equivalent resistance will be approximately $1/g_{m2}$ at the node 3. Due to Miller effect, effective capacitance of C_{ds2} seen at node 2 is 300 times higher than C_{gs2}. In this case, we can neglect C_{gs2}.
Adding a big capacitor between the gate and drain also pushes the zero emerged in the second stage towards lower frequencies. That is why we need to take the zero into account for our circuit analysis at this point. The zero is on the right half-plane, so it will produce phase lag.

New poles:
$w_{p1} = \frac{1}{R_S*(C_{gs1}+C_{gd1}*(1+g_{m1}*R_1))} = \frac{1}{500*130f} = 15.4Grad/s$
$w_{p2} = \frac{1}{R_1*(C_c*g_{m2}*R_L)} = \frac{1}{5k*10*1p} = 20Mrad/s$
$w_{p3} = \frac{g_{m2}}{C_L} = \frac{20m}{1.1pF} = 18.2Grad/s$
$|z_1| = w_{z1} = |\frac{g_{m2}}{C_c}| = \frac{20m}{1pF} = 20Grad/s$

Total phase shift= $tan^{-1}\frac{2G}{20M} + tan^{-1}\frac{2G}{15.4G} + tan^{-1}\frac{2G}{18.2G} + tan^{-1}\frac{2G}{20G} = 108.8^o$
Phase margin $= 180^o - 108.8^o = 71.2^o$

Exercise 3

a) After adding series resistance to C_c, new zero will be:
$|z_1| = w_{z1} = |\frac{1}{(\frac{1}{g_{m2}} - R) * C_c}|$

The frequency of second non-dominant pole is 15.4Grad/s
So, $w_{z1} = \omega_{p1} = 15.4Grad/s = |\frac{1}{(\frac{1}{g_{m2}} - R) * C_c}|$
$=> R = 64.9\Omega$

b) Total phase shift$= tan^{-1}\frac{2G}{20M} + tan^{-1}\frac{2G}{18.2G} = 95.7^o$
Phase margin $= 180^o - 95.7^o = 84.3^o$